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# Andrey Popov

AI Solutions Engineer | Agentic workflows & MCP | Data & ML pipelines | Finance background

Neuchâtel, Switzerland · cv.andrey.popov@gmail.com · vospr.github.io · linkedin.com/in/andrey-p-58ab10100

## PROFILE

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My background was in making financial numbers provable. Now I work between businesses facing AI challenges and the implementation of AI solutions in their processes. I frame the problem, decide what is safe to automate, design how the result gets verified, and direct AI coding agents to build it. I own the design, the validation and the decisions, and every claim I make about my work links to a test, a log or a receipt. You can find my latest project at [vospr.github.io](https://vospr.github.io).

## EXPERIENCE

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**EPAM Systems · Sep 2024 – present**

**Software Engineer, primary skill AI Solution Engineering**

› *AI coaching platform for a leadership-coaching company (client engagement) · May 2026 – present*

- **One-day MCP decision memo that opened the engagement:** could the coaching engine reach employees inside enterprise AI assistants through one MCP server, and what is safe to expose before identity and GDPR work? I recommended stateless single-turn tools now and multi-turn coaching later; the July 2026 MCP specification later made stateless the protocol default.
- **One of ~6 engineers** on an AI-assisted team rebuilding the client's coaching assistant (web and Slack; internal pilot in production, external launch planned). The architecture is the tech lead's. My share on the team board: I'm second of 11 contributors (16% of closed, assigned issues and 8% of merged pull requests).
- **Release readiness:** wrote the deployment, QA and go/no-go plans for the first internal milestone, then built the controls behind them: a kill switch for AI generation and notifications, a cross-tenant isolation test suite, a CI check that every user-content column is encrypted, deploy-time smoke tests, and a CI guard that catches pull requests silently reverting merged work.
- **Owner of the human-coaching integration (TypeScript):** service-to-service auth to a separate coaching-operations platform, coach matching, booking and cancellation with consistent calendar slots, a video-call screen (Azure Communication Services), and cross-system admin tools.
- **AI delivery-cost measurement:** designed and built the cost track of an internal AI-first programme, running monthly since Aug 2026. A change counts only when it merges to main; spend is reported per change as a distribution. I found and fixed counting errors in both tools' logs before publishing. One breakdown traced ~90% of a ticket's tokens to a single never-reset session.

› *Internal candidate–position ranking system (team-owned, in production) · Mar – May 2026*

- **One engineer on a production ranking pipeline:** ~400 dbt models on BigQuery filter tens of millions of candidate–position pairs per run before a multi-task PyTorch model ranks them. My scope: matching rules, a pipeline refactor, a feature ablation. Productive on the 1,400-file codebase in about a week.
- **Behaviour-preserving refactor:** restructured the forecast and training pipelines behind a parity safety net: compile-diff, row-level parity on the final tables, and an outcome check showing 99.8% ranking-bucket stability. Consolidated ~900 duplicated SQL lines into shared macros and added 148 dbt tests where there had been none.
- **Business rules and ablation:** three candidate-exclusion rules as NULL-safe SQL predicates, sized by a 100k-pair simulation (validated on real data, not yet live when I left). A permutation-plus-retrain ablation supported retiring a hand-built location feature.

*AI-native engineering track · Nov 2025 – Mar 2026:* personal projects below, built with AI coding agents under my specification and review.

- **Governed AI decision workflow: Verdaca ([github.com/verdaca/VERDACA](https://github.com/verdaca/VERDACA)).** A producer, an isolated reviewer and a synthesizer deliberate over real model calls; every run emits a JSON receipt with evidence, critique, per-call cost and stop reasons (each recorded run about \$0.03), including one run from Claude Desktop over MCP. Ports-and-adapters design; 440+ root tests and 1,450+ kernel tests in CI. Known limits are documented.
- **MLOps: Seismic facies classification pipeline.** Rebuilt on the public F3 benchmark: one sklearn Pipeline served as-is through MLflow and FastAPI, a promotion gate that must beat a baseline and the current champion, leak-free train/validation/test split. Test accuracy 0.72 vs 0.51 baseline.
- **LLM evaluation: LangGraph concierge routing.** Rules-then-LLM routing measured on 40 labelled queries: 52% rules-only, 65% with Claude Haiku vs 62% with Opus at about a third of the cost, so the router defaults to Haiku.
- **Context engineering: Claude Code agent template.** Dispatcher, role subagents, blind review, file memory and hooks verified blocking a live session. Presented at an internal EPAM workshop (~40 engineers).

**Université de Neuchâtel · MSc student · 2023 – 2026**

Time-series forecasting for the MSc thesis: next-day hourly Swiss electricity spot prices from regression, ARX and neural-network models and their combination, tested out of period on 2024 data.

## Earlier roles: finance, reporting and analytics (2005 – 2025)

- **Novartis Baltics · Manager Financial Reporting & Accounting · Aug 2024 – Jan 2025:** statutory closing and Baltic reporting review (Latvia, Lithuania, Estonia); SOX 302/404 controls; SAP FI/CO controlling and financial-reporting support.
- **Forevers (food manufacturing) · Data Analyst · Nov 2021 – Aug 2024:** FP&A, product costing and margin analysis; led the redesign of the planned-vs-actual cost model and its finance data pipeline.
- **X Infotech (digital ID platforms) · Business Analyst · May 2018 – Oct 2021:** monthly close, IFRS accruals, reconciliations and subgroup consolidation; automated finance data controls.
- **Tele2, Stockholm · Business Analysis Team Lead, Revenue Management · Dec 2014 – Mar 2018:** pricing, demand forecasting and channel strategy.
- **Greenatom, Rosatom group finance shared-service centre · Consolidated Financial Reporting Manager · Jan 2009 – Jun 2014:** IFRS consolidated statements, including SAP implementation and migration projects; led a team of 35.
- **Grant Thornton · Consulting Team Lead · Jan 2005 – Jan 2009:** audit and consulting.

## EDUCATION

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**MSc in Applied Economics, specialization Data Science**, University of Neuchâtel · completed Sep 2026

Thesis: *Combining Machine Learning with Traditional Forecasts: Swiss Electricity Spot Prices* (grade 5/6)

**MBA**, Stockholm School of Economics in Riga · 2015

**MSc in Finance, Banking and Corporate Finance**, Lomonosov Moscow State University (MSU) · 2000

## CERTIFICATIONS

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- **Anthropic:** Claude Certified Architect – Professional (2026) · Claude Certified Architect – Foundations (2026)
- **AWS:** Certified Generative AI Developer – Professional (2026) · Certified AI Practitioner (2026)
- **Databricks:** Generative AI Engineer Associate (2025, valid to 2027) · Data Engineer Professional (2025, valid to 2027)
- **Microsoft:** Azure Data Scientist Associate, DP-100 (2025) · Fabric Analytics Engineer Associate, DP-600 (2025)
- **Google:** Gemini Enterprise Agent Development, Partner Specialist (2026)
- **ACCA:** Diploma in International Financial Reporting (DipIFR)

## SKILLS

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Agentic workflows with Claude (multi-agent loops, JSON receipts, MCP servers) · LLM evaluation (labelled eval sets, model-cost comparison) · LLM gateway patterns (LiteLLM, budgets, cost metering) · AI delivery-cost measurement · Python (pandas, scikit-learn, statsmodels) · SQL · dbt · BigQuery · ML validation (leak-free splits, baselines, promotion gates) · econometrics and forecasting · MLflow · FastAPI · GitHub Actions CI · TypeScript / Node · Azure · GCP · PyTorch · LangGraph · Databricks · Spark · Microsoft Fabric · React · Docker

## STRENGTHS

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Analytical mindset · Problem framing and scoping · Validation design (parity gates, out-of-sample tests, baselines) · Finance domain depth · Fast ramp-up on unfamiliar systems · AI-assisted delivery with measured cost

## LANGUAGES

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English (fluent) · Latvian (B1) · French (A2) · Russian (native)